

Advances in Superconducting Undulators

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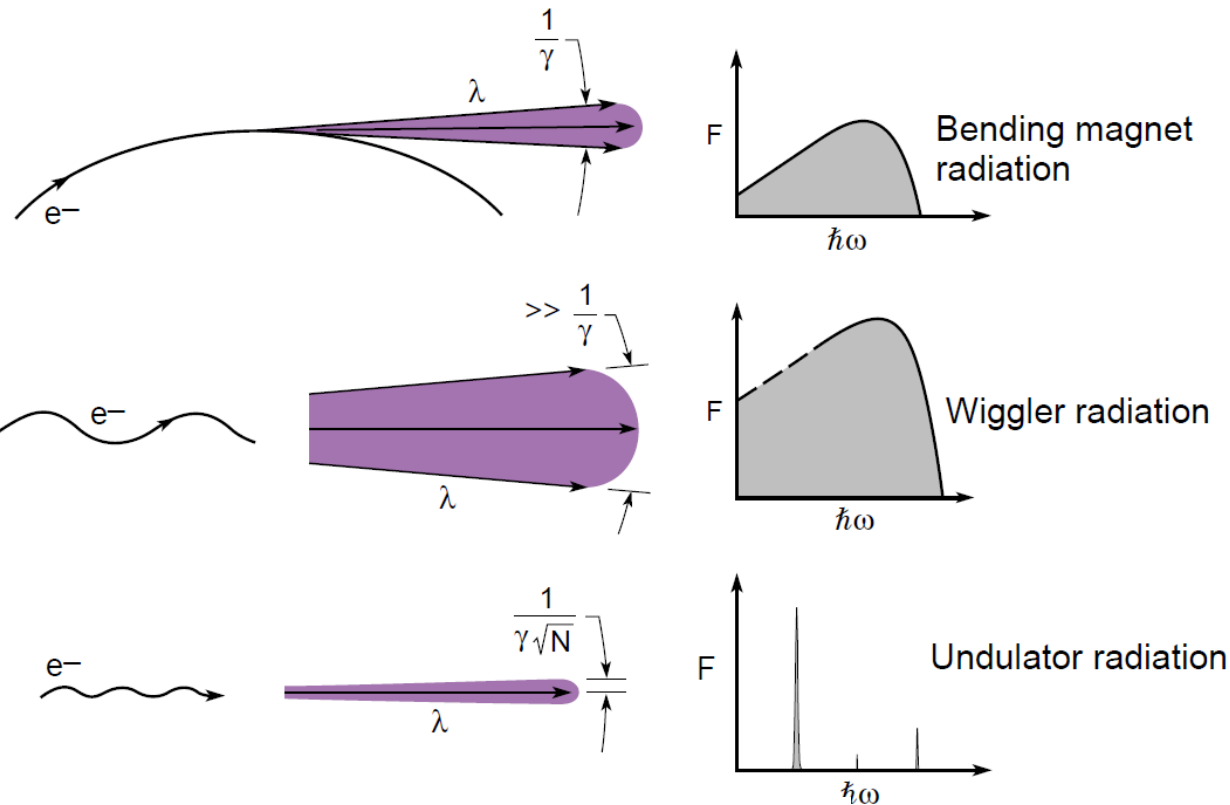
NA-PAC'13, Pasadena, California, October 4, 2013

Work supported by U.S. Department of Energy, Office of Science,
under Contract No. DE-AC02-06CH11357.

Scope

- What is a superconducting undulator (SCU) ?
- Motivation for superconducting undulators?
- SCU challenges
- Solutions:
 - Magnet
 - Magnetic shimming
 - Heat loads and cooling
 - Measurement system
- Experience with SCU operation
- Activities in SCUs
- Helical SCUs
- Conclusions

Forms of synchrotron radiation



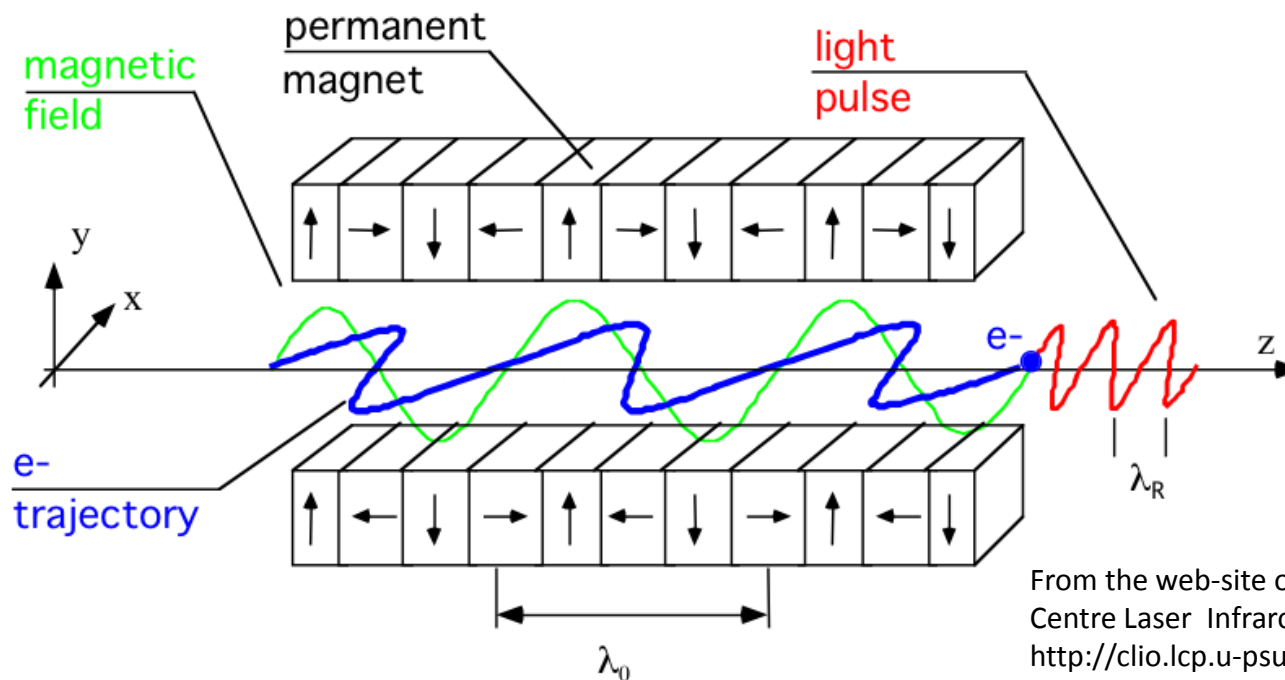
Undulator radiation wavelength and photon energy:

$$\lambda = \frac{\lambda_u}{2\gamma^2} \left(1 + \frac{K^2}{2} + \gamma^2\theta^2 \right) \quad E(\text{keV}) = \frac{0.9496 E_e^2(\text{GeV})}{\lambda_u(\text{cm}) \left(1 + \frac{K^2}{2} + \gamma^2\theta^2 \right)}$$

$$\text{where } K \equiv \frac{eB_0\lambda_u}{2\pi mc} = 0.9337 B_0(\text{T})\lambda_u(\text{cm})$$

From lectures by Prof. David T. Attwood, <http://ast.coe.berkeley.edu/sxreuv/>

Undulator radiation



From the web-site of
Centre Laser Infrarouge d'Orsay:
http://clio.lcp.u-psud.fr/clio_eng/FELrad.html

In coordinate frame that moves with an electron in Z:

Electron 'sees' the magnetic structure with the period length λ_0/γ moving towards it, and emits as a dipole at the wavelength $\lambda^* = \lambda_0/\gamma$, where γ is the relativistic Lorentz factor.

In laboratory (observer) frame:

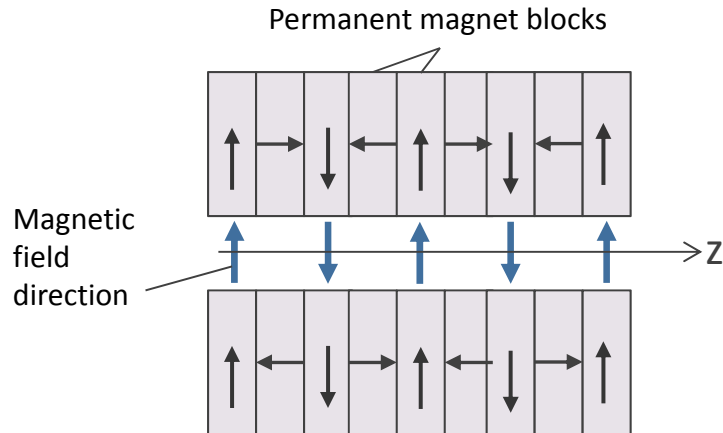
Observer sees this dipole radiation shifted to even shorter wavelength, through the relativistic Doppler effect.

In the forward direction, the observed wavelength of the radiation is $\lambda_R = \lambda^* \gamma(1-\beta) = \lambda_0(1-\beta) = \lambda_0/2\gamma^2$.

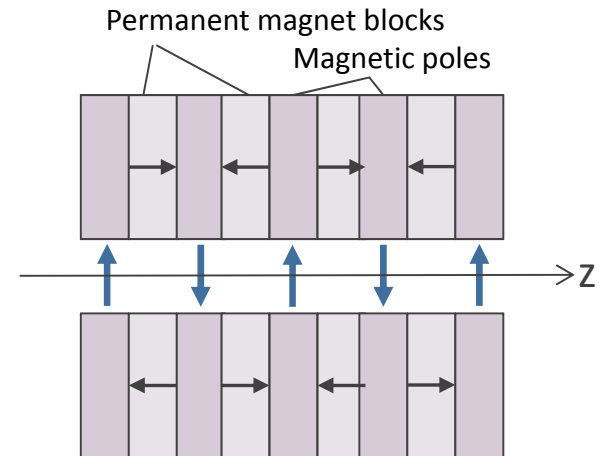
As a result, a 3.3-cm undulator can emit 10-keV photons on a 7-GeV electron storage ring ($\gamma = 13700$).

Undulator magnetic structures

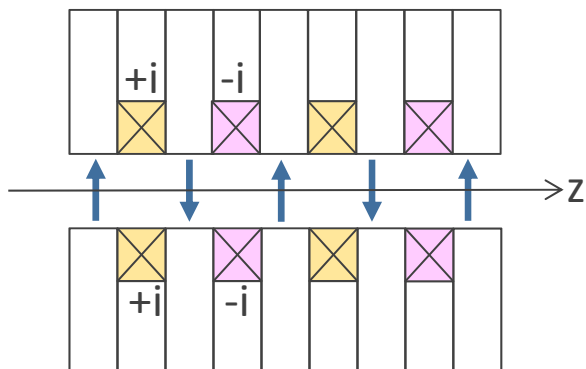
Permanent magnet structure



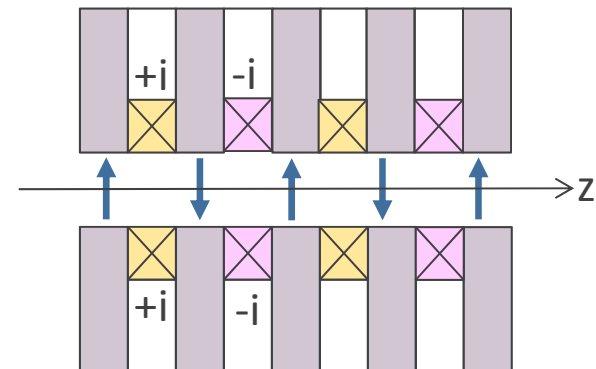
Hybrid structure



Electromagnet structure



**Electromagnet structure
with magnetic poles**



Scope:

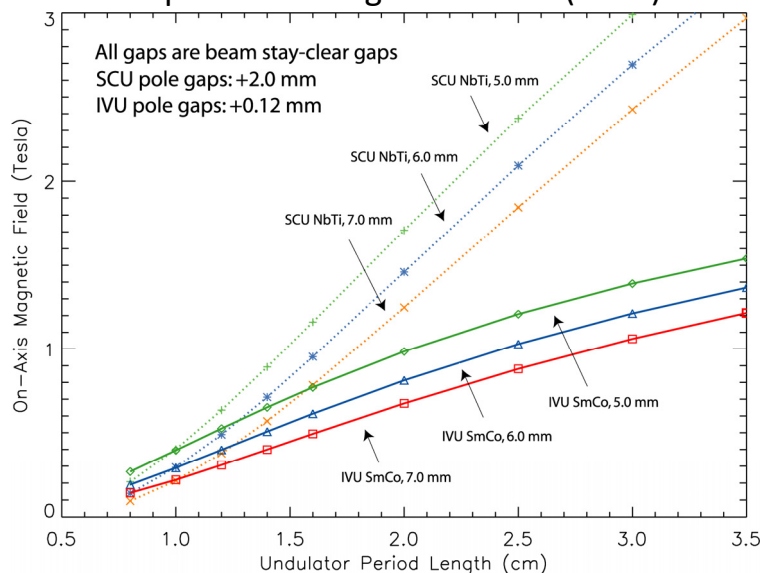
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SCU motivation - Higher undulator field

	CPMU PrFeB	SCU NbTi	SCU NbTi-APC
Undulator period, mm	15	15	15
Pole gap, mm	5.2	6	6
Undulator field, T	1.0	1.18	1.46
Undulator parameter K	1.40	1.65	2.05

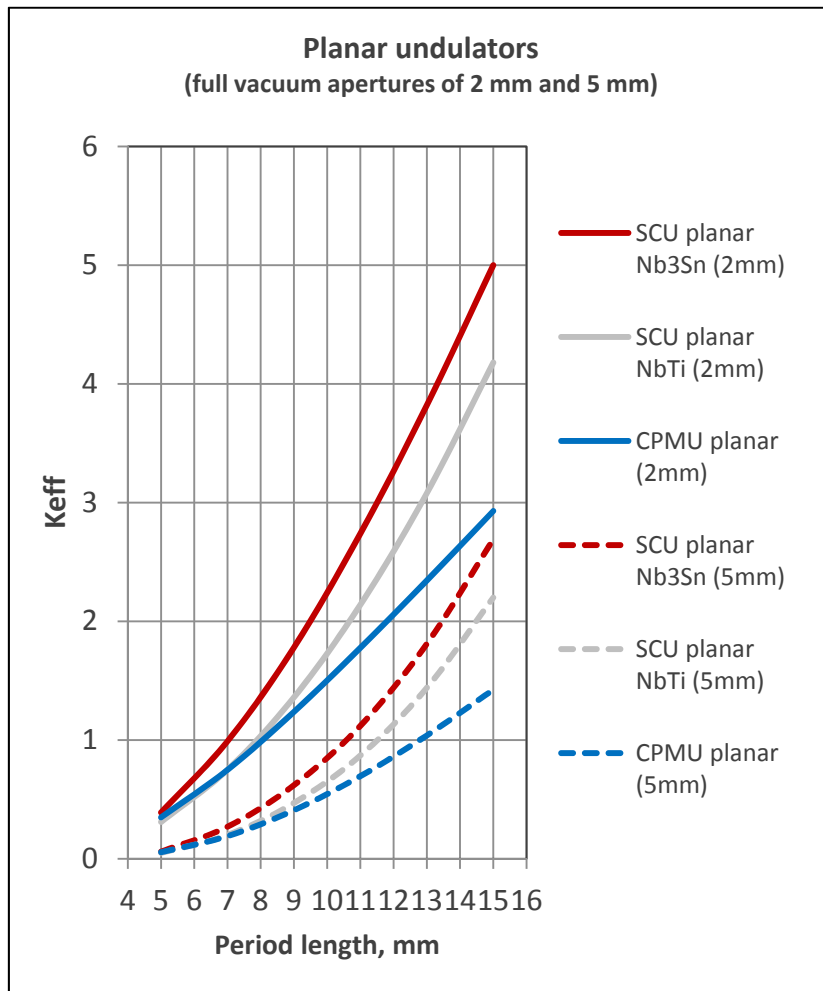
S. Casalbuoni et al., "Test of short mockups for optimization of superconducting undulator coils," presented at MT23, Boston, 2013.

Magnetic fields for in-vacuum undulators (IVUs) and superconducting undulators (SCUs)



R. Dejus et al., "On-Axis Brilliance and Power of In-Vacuum Undulators for the Advanced Photon Source," MD-TN-2009-004

Y. Ivanyushenkov, "Advances in Superconducting Undulators", NA-PAC'13, October 4, 2013



J. Bahrtdt and Y. Ivanyushenkov, "Short Period Undulators for Storage Rings and Free Electron Lasers," Journal of Physics: Conference Series 425 (2013) 032001.

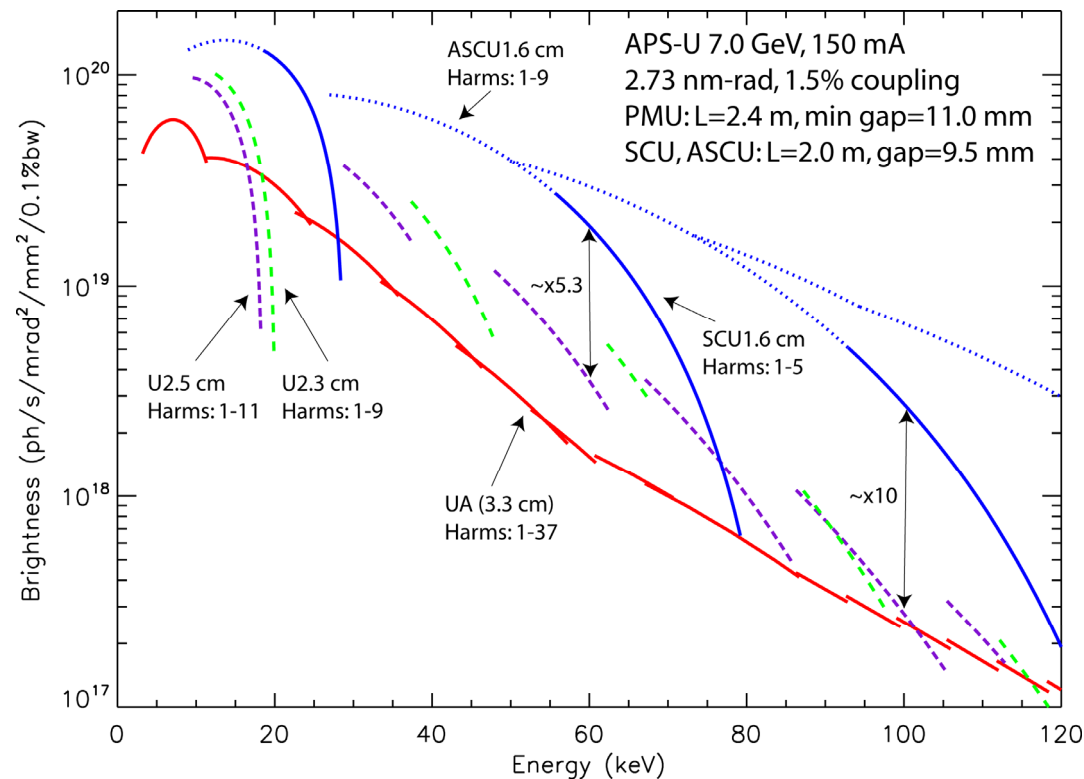
SCU motivation - Higher photon fluxes

Superconducting planar undulator at 3-GeV Diamond Light Source(DLS) will increase photon flux of ~ 15 times and brightness of ~ 20 times at 40 keV when compared to the current in-vacuum undulator [1].

[1] J. Clarke et al., "Status of the UL Superconducting Planar Undulator Project," Proc. of IPAC2013, WEPWA062, p. 2259.

Advanced Photon Source (APS) undulator brightness tuning curves [2]

(SCUs 1.6 cm vs. UA 3.3 cm vs. Revolver U2.3 cm & U2.5 cm)



[2] R. Dejus, private communication.

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SCU challenges

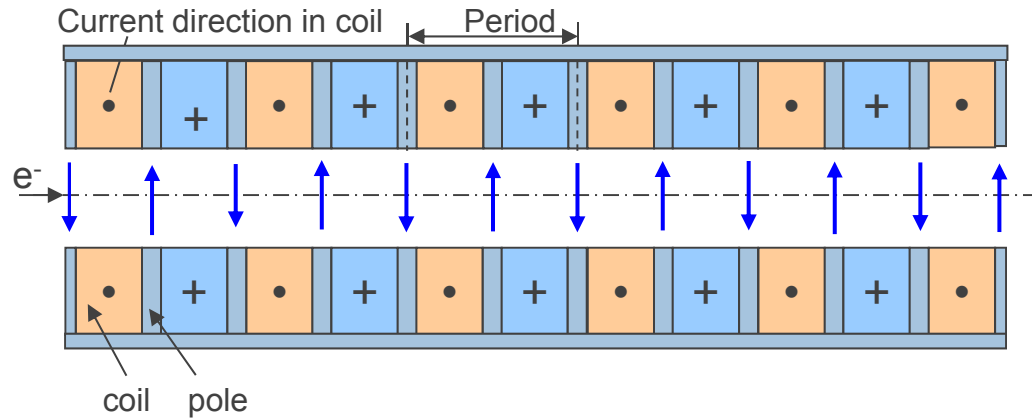
SCU as a superconducting magnet	SCU as an insertion device	SCU as a photon source
- Choice of superconductor; - Design and fabrication of magnetic structure; - Cooling of superconducting coils in presence of beam heat load; - Design and fabrication of SCU cryomodule.	- Low field integrals; - Measurement of SCU performance before installation into storage ring.	- High quality field: • Trajectory straightness; • Low phase error. - Shimming technique.

Scope:

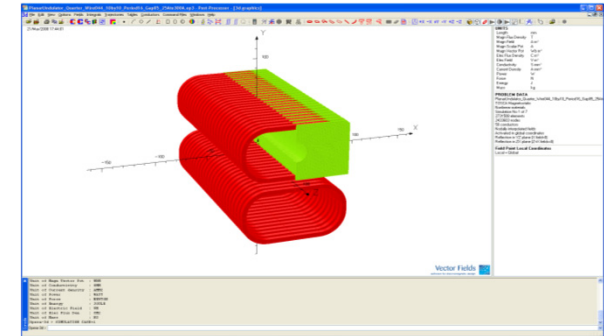
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SCU magnet - Low temperature superconductor (LTS) coils

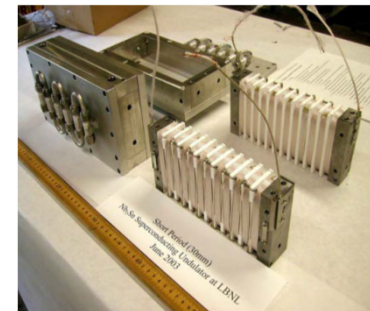
Current directions in a planar undulator



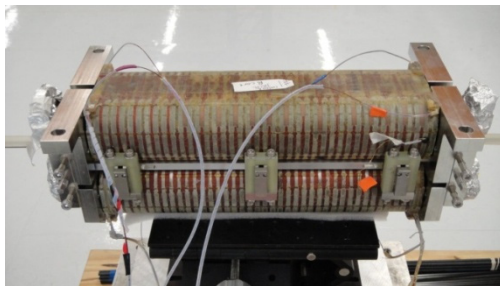
Planar undulator winding scheme



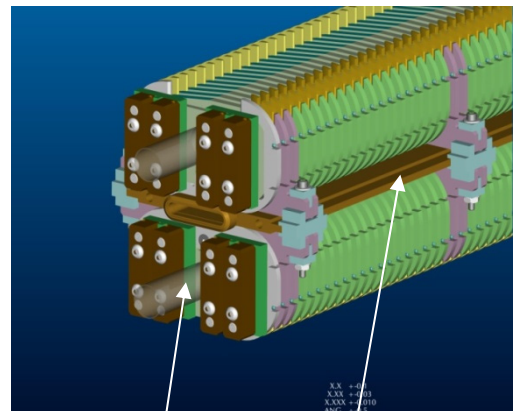
Nb₃Sn SCU magnet prototypes at LBNL



NbTi SCU0 magnet at APS



Magnetic structure layout



LHe cooling channel

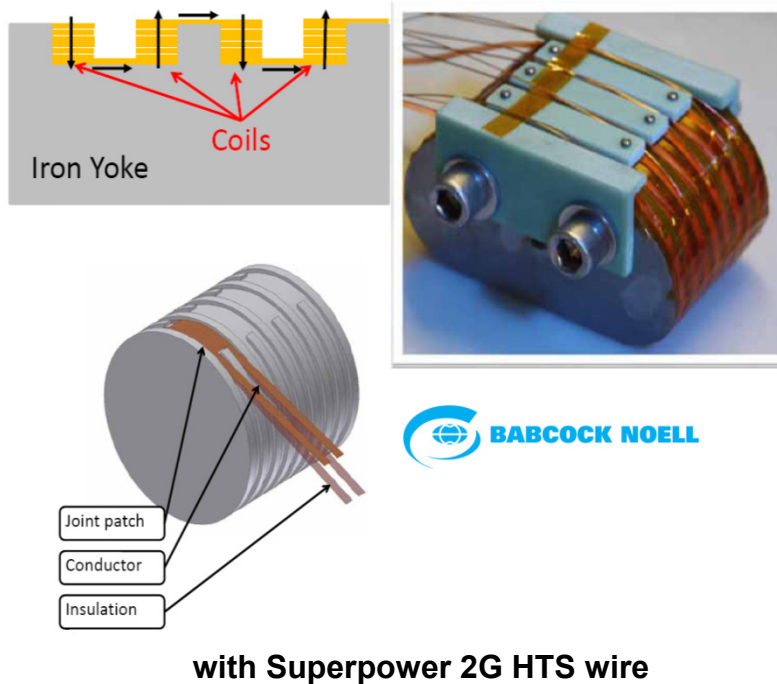
Beam vacuum chamber



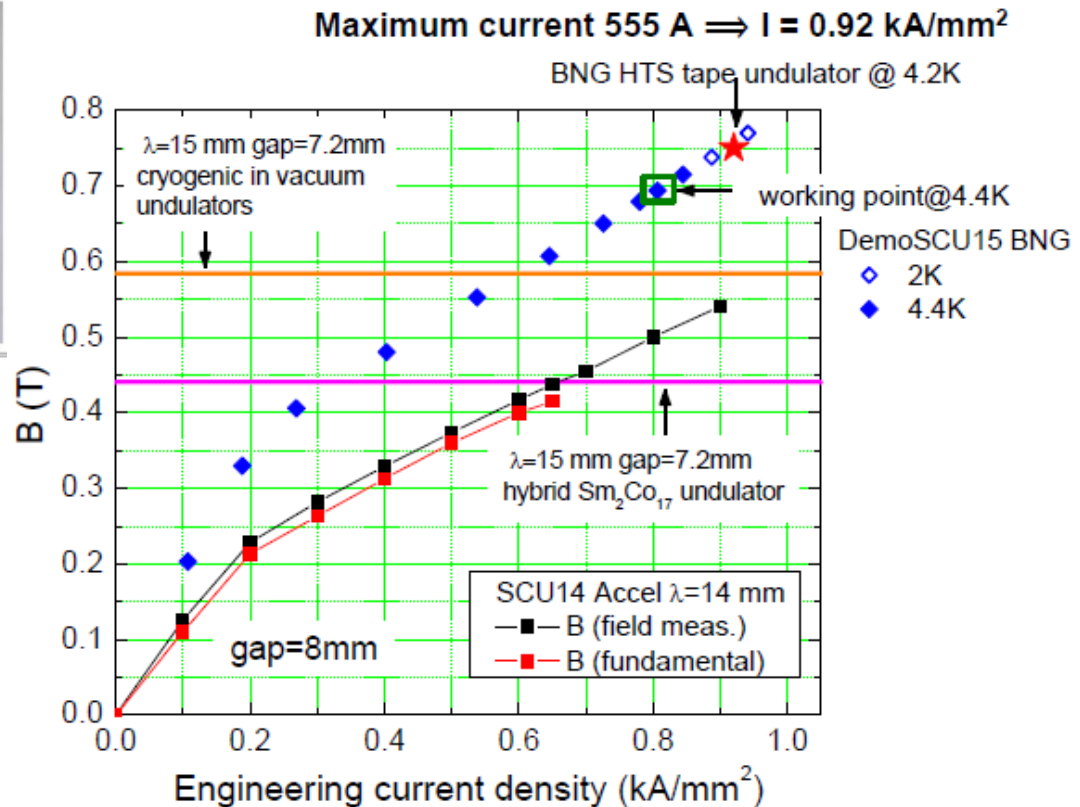
S. Prestemon et al., "Superconducting Undulators for the Next Generation of Synchrotron Radiation Sources," presented at LBNL Interdisciplinary Instrumentation Colloquium, July 2006.

SCU magnet - High temperature superconductor (HTS) coils

HTS tape planar undulator



HTS tape planar undulator mockup:
results of test at CASPERI (ANKA, KIT) [1]



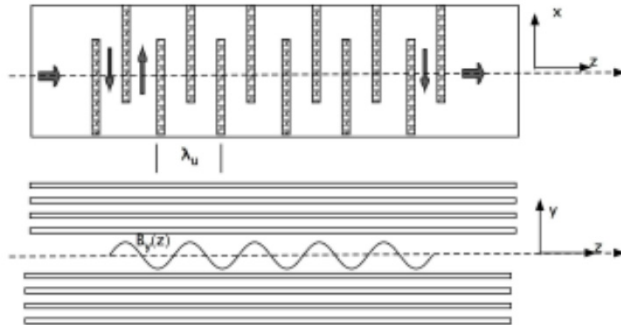
C. Boffo, IDMAX10

<http://www.maxlab.lu.se/usermeeting/2010/sessions/>

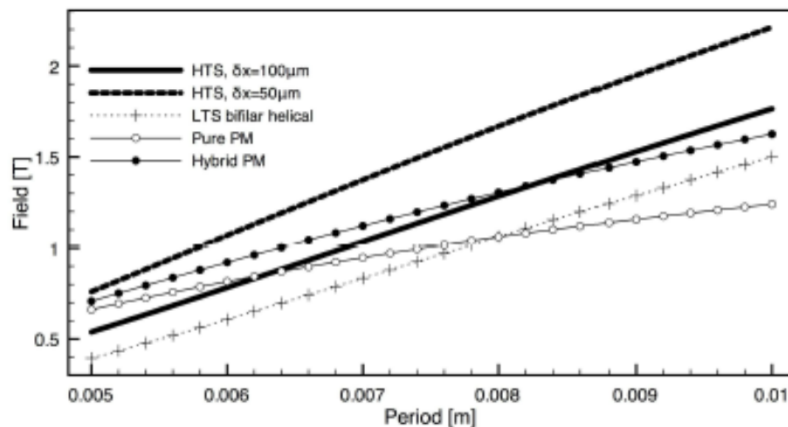
[1] S. Casalbuoni, private communication.

SCU magnet - HTS tape undulator

Basic concept of the tape undulator [1]



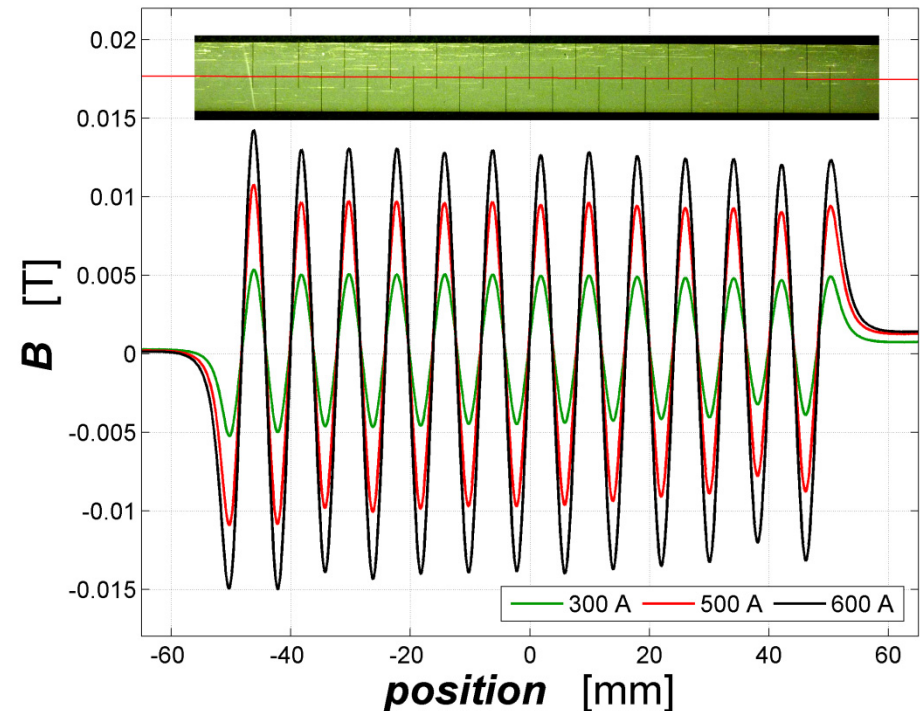
Performance comparison between PM, PM hybrid, superconducting helical, and the proposed HTS tape concept technologies.



[1] S. Prestemon et al., "High Performance Short-Period Undulators Using High Temperature Superconductor Tapes," Proc. of PAC09, WE5RFP075, p. 2438 (2010).

First tests on laser structured wire at ANKA [2]

Etching using Trumpf picosec YAG - IR laser, programmable beam control used for Roebel cables



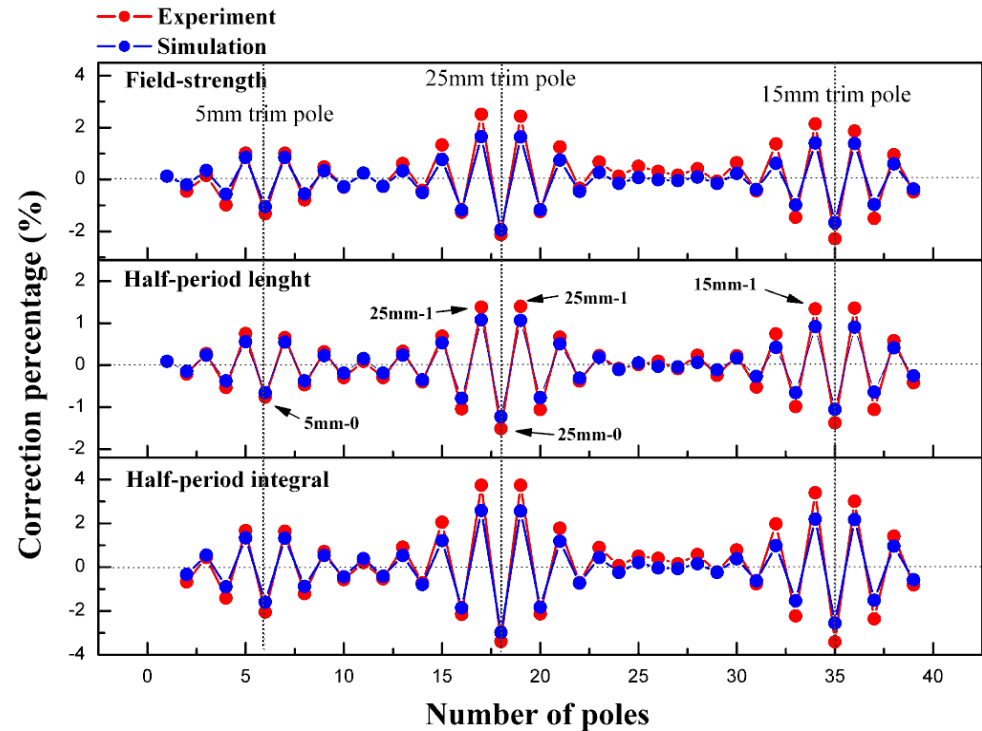
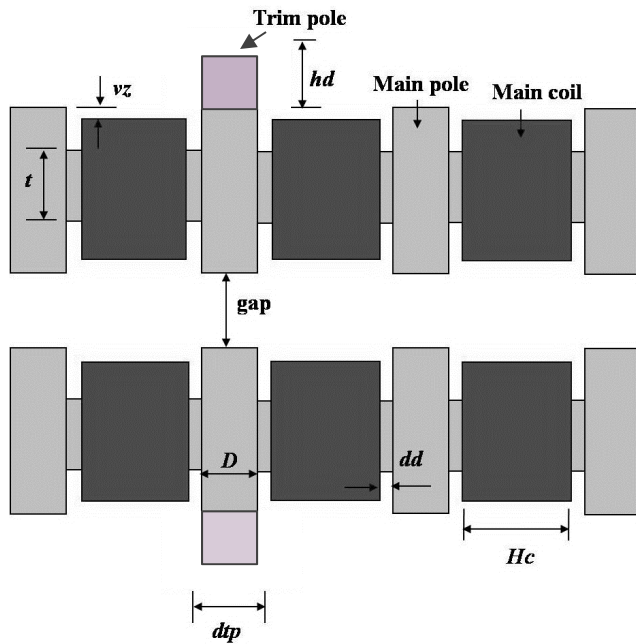
KIT internal collaboration:
ANKA with ITEP (W. Goldacker & R. Nast)

[2] S. Casalbuoni, private communication.

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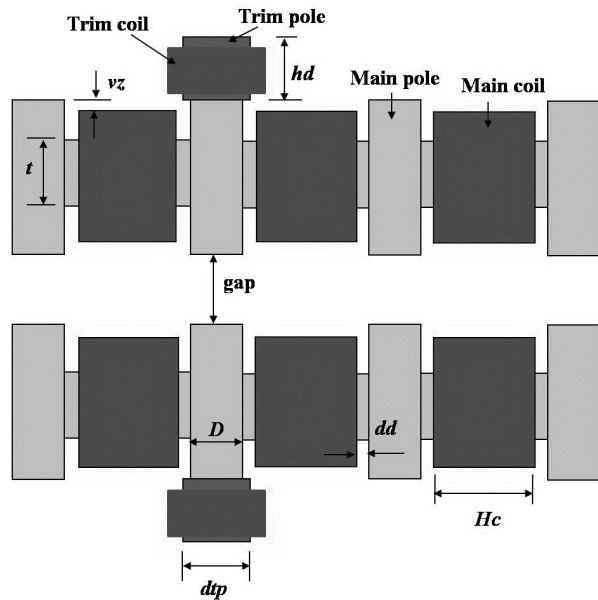
SCU shimming - Additional magnetic material



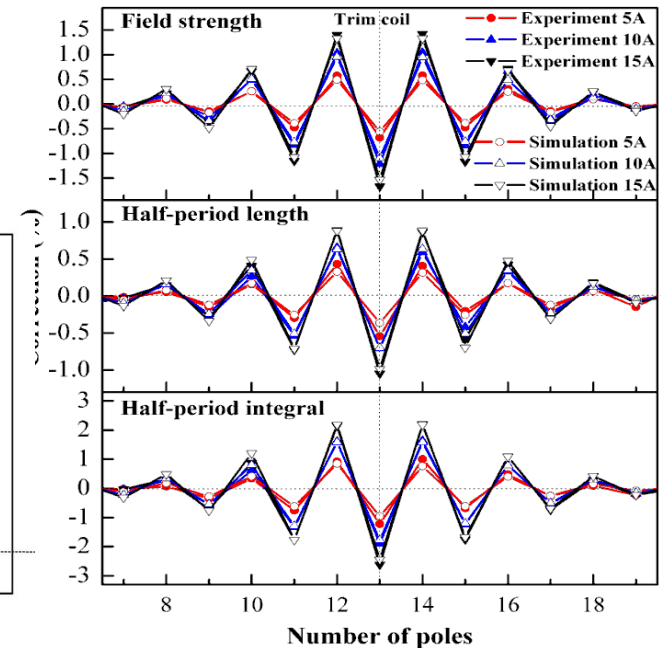
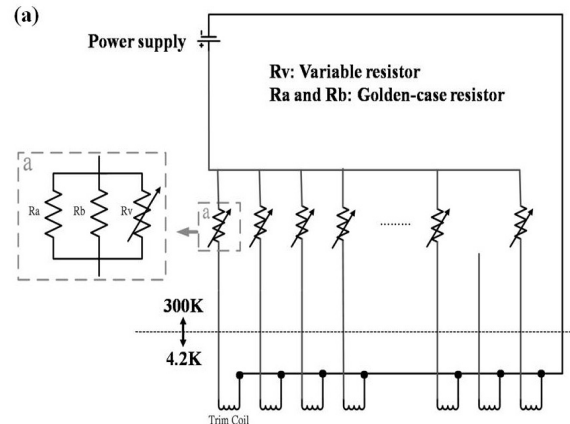
- Field profile is first measured
- Thin iron pieces are then mounted on the poles where the correction is required
- **Correction effect:** about 2%
- **Pros:** no extra heat load;
- **Cons:** long warm-up and cold-down time

J.-C. Jan et al., "Field Trimming by Iron Pieces and Coils in a Superconducting Undulator," IEEE Trans. Appl. Superconductivity. **19**, 1332 (2009).

SCU shimming - Trimming coils



(a)

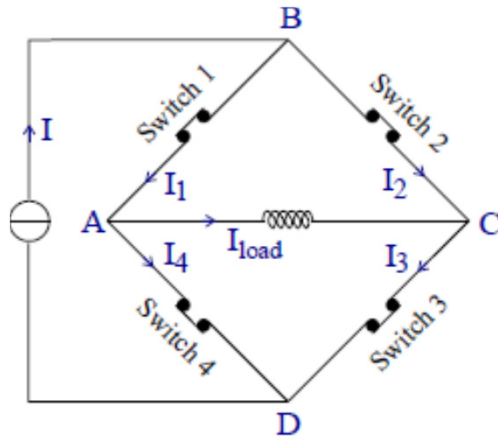


- Trim coils are mounted on trim poles
- Trim coil currents are adjusted by variable resistors
- **Correction effect:**
-0.7% (5 A - 0), -1.2%(10 A - 0), -1.7 % (15 A - 0)
- **Pros:** in-situ field shimming
- **Cons:** addition heat load through many current leads

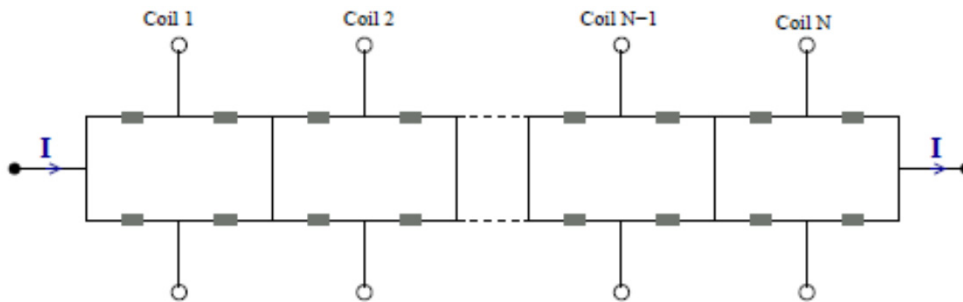
J.-C. Jan et al., "Field Trimming by Iron Pieces and Coils in a Superconducting Undulator," IEEE Trans. Appl. Superconductivity. **19**, 1332 (2009).

Trimming coils with superconducting switches

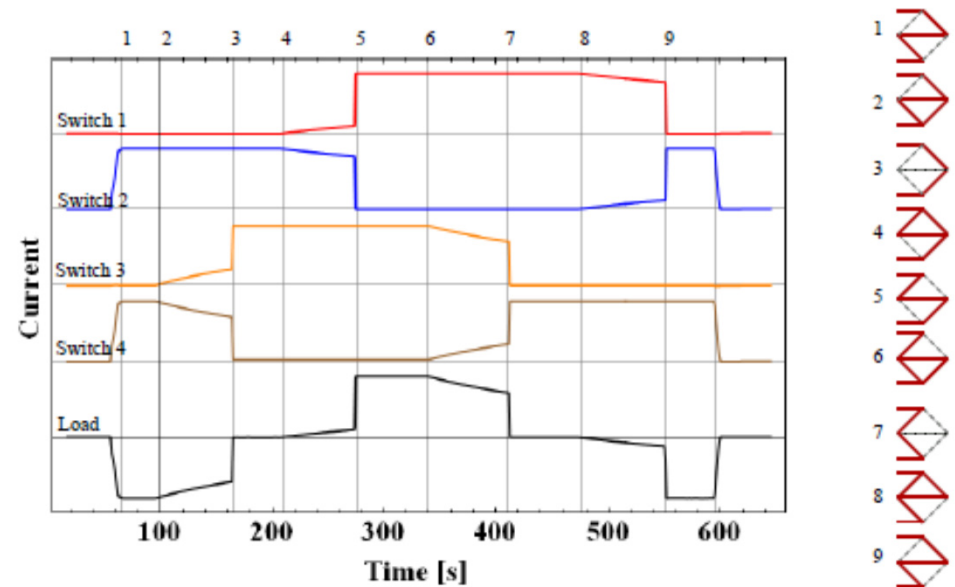
Superconducting bridge concept



Extended superconducting bridge concept to control N coils



Experimental demonstration of superconducting bridge operation

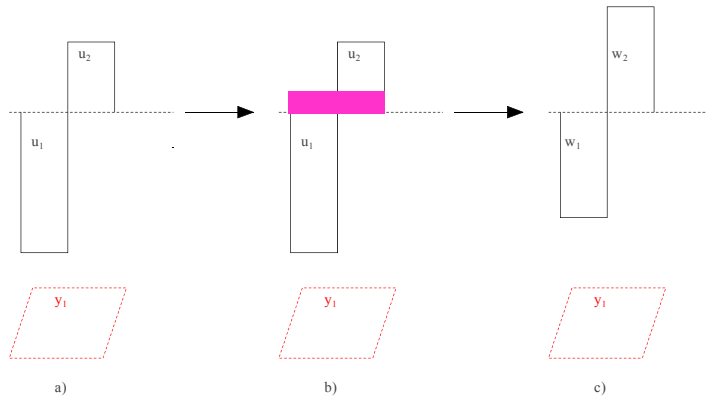


Left: Current vs. time in the five branches of the circuit
Right: Current path in the bridge circuit

A. Madur et al., "Superconducting switch concept applied to superconducting undulator phase error correction," AIP Conf. Proc. 1234, 552 (2010).

Passive induction shimming - A concept

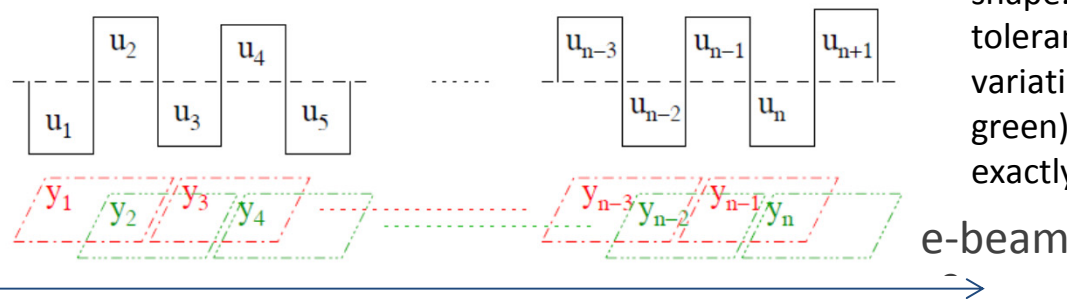
Concept of induction shimming – Single superconducting loop



Faraday's law of induction: a change of the magnetic flux through the closed loop is compensated by the magnetic flux produced by the induction current.

A superconducting loop covers a single undulator period. Without the induced current in the loop (a) the magnetic flux in the two half periods is different. For the sake of simplicity it is assumed that the flux has a rectangular shape. When the loop is installed, the induced current produces a correction flux (red in b) and equalizes the field strength.

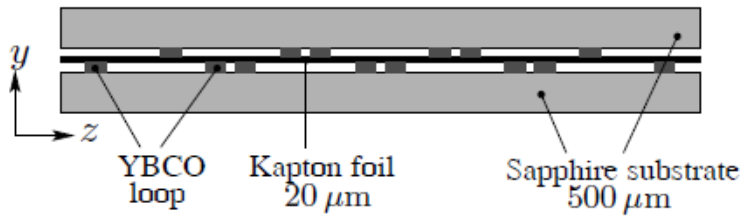
Concept of induction shimming – Multiple superconducting loops



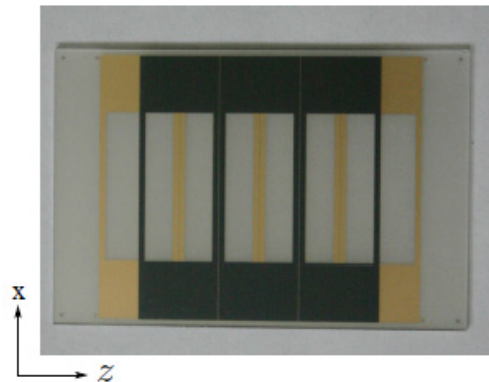
The sinusoidal field in one of the undulator-halves is approximated for the sake of simplicity by a rectangular shape. Due to minor field errors caused by finite tolerances the amplitude of the field varies slightly. This variation is compensated by HTSC loops (red and green). The length of one loop in the beam direction is exactly one ideal period length.

D. Wollmann et al., "A New Concept for Reducing Phase Errors in Superconducting Undulators: Induction-Shimming," Proc. of EPAC08, p. 2323.

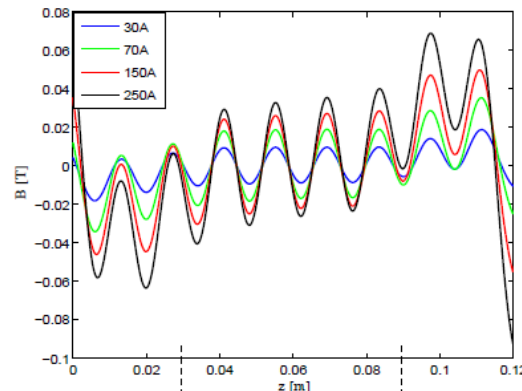
Passive induction shimming - Experimental demonstration



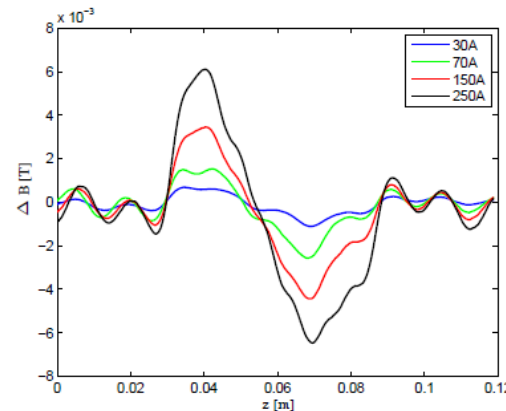
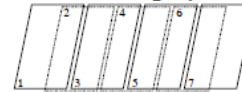
Cut through the stacked induction-shimming structure with the substrate. The YBCO loops are arranged on a 500 μm thick sapphire substrate. The thickness of the YBCO layer is 330 nm. The period of the mock-up is 14mm.



Photograph of the set of seven correction loops. Two 500 μm thick sapphire substrates are sputtered with four and three YBCO loops with a thickness of 330 nm, covered with a 200 nm thick gold layer. Dimensions of the loops: 14mm $\times$ 44mm. The two substrates are stacked and mounted at a distance of 1mm from the surface of the undulator.



closed-loop system



The influence of the YBCO closed loop system on the field shown in Fig. 2 for various undulator currents (above) and the difference of the field measurements with and without the YBCO loops system (below). The YBCO loop system tries to compensate the field which increases from left to right

D. Wollmann et al., "Experimental Demonstration of the Induction Shimming Concept in Superconducting Undulators," Proc. of PAC09, p. 2474.

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SCU heat loads

Estimated heat load on 4.2 K beam duct and cryostat, W [1]

	Beam duct	Cryo-stat
Conduction	0.4	0.95
Thermal radiation	0.08	0.08
Synchrotron radiation	0.1	0
Image current at 400 mA	4.3	0
Total:	5.46	

[1] C.S. Hwang et al., "Mini-pole Superconducting Undulator for X-Ray Synchrotron Light Source," IEEE Trans. Appl. Supercond. 16(2) (June 2006) 1855.

Static and beam-induced heat loads on SCU components, W [2]

Heat Source	4 K	20 K	60 K
Electron beam		10	6
Thermal radiation	0.017	0.205	0.3
Conduction through:			
- beam tube bellows			4.8
- beam tube support at 20 K	0.08		
- He vent bellows	0.0004	0.008	0.6
- cold mass support	0.08		
- thermal shield supports		0.22	3.25
Main current leads	0.285		47
Correction current leads	0.064		8.02
Total	0.526	10.43	69.97

[2] Y. Shiroyanagi et al., "Thermal Modeling of the Prototype Superconducting Undulator (SCU0)," THPAC07, this conference.

SCU beam-induced heat loads

Beam-induced heat loads on SCU beam chamber

Continuous wave (CW) heat loads

Heat Source	Value, W [1]	Value, W [2]
Image current		4.7
Wakefields	~40	< 0.5
Synchrotron radiation from upstream dipole		0.2
Injection losses		0.1-2
Beam lifetime losses		<< 1
Electron cloud		< 2
Total:		≈ 10

Transient heat loads

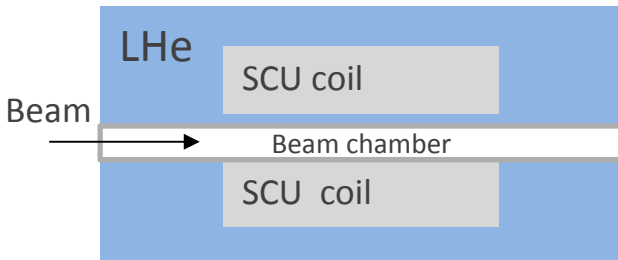
Heat Source	Value, W [2]
Injected beam loss	13
Corrector failure	20
Synchrotron radiation with steering errors	25
Electron beam steering errors	20
Max. transient heat load	25

[1] J. Clarke et al., "Status of the UK Superconducting Planar Undulator Project", Proc. of IPAC2013, p. 2259.

[2] K. Harkay et al., "Beam-Induced Heat Load Predictions and Measurements in the APS Superconducting Undulator," WEPSM06, this conference.

SCU cooling schemes

Direct cooling of SCU coils



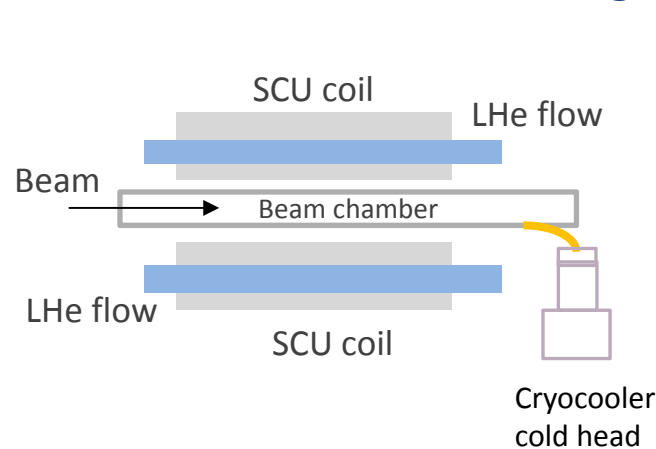
Pros:

- SCU coils is direct contact with LHe

Cons:

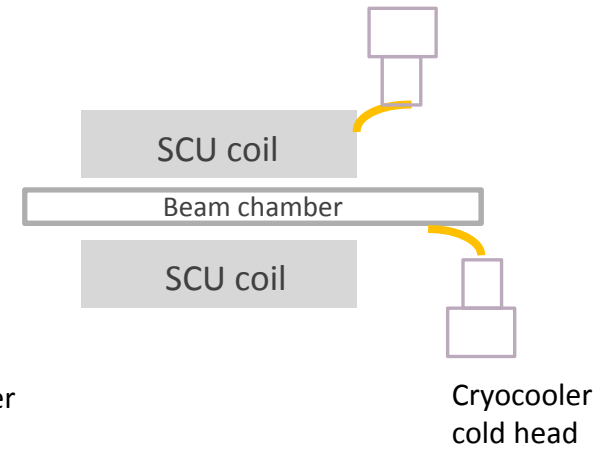
- Beam heats LHe

Indirect cooling of SCU coils



Pros:

- No heating of LHe by beam
- ### Cons:
- Possible temperature difference between the LHe and the coil;
 - LHe pump



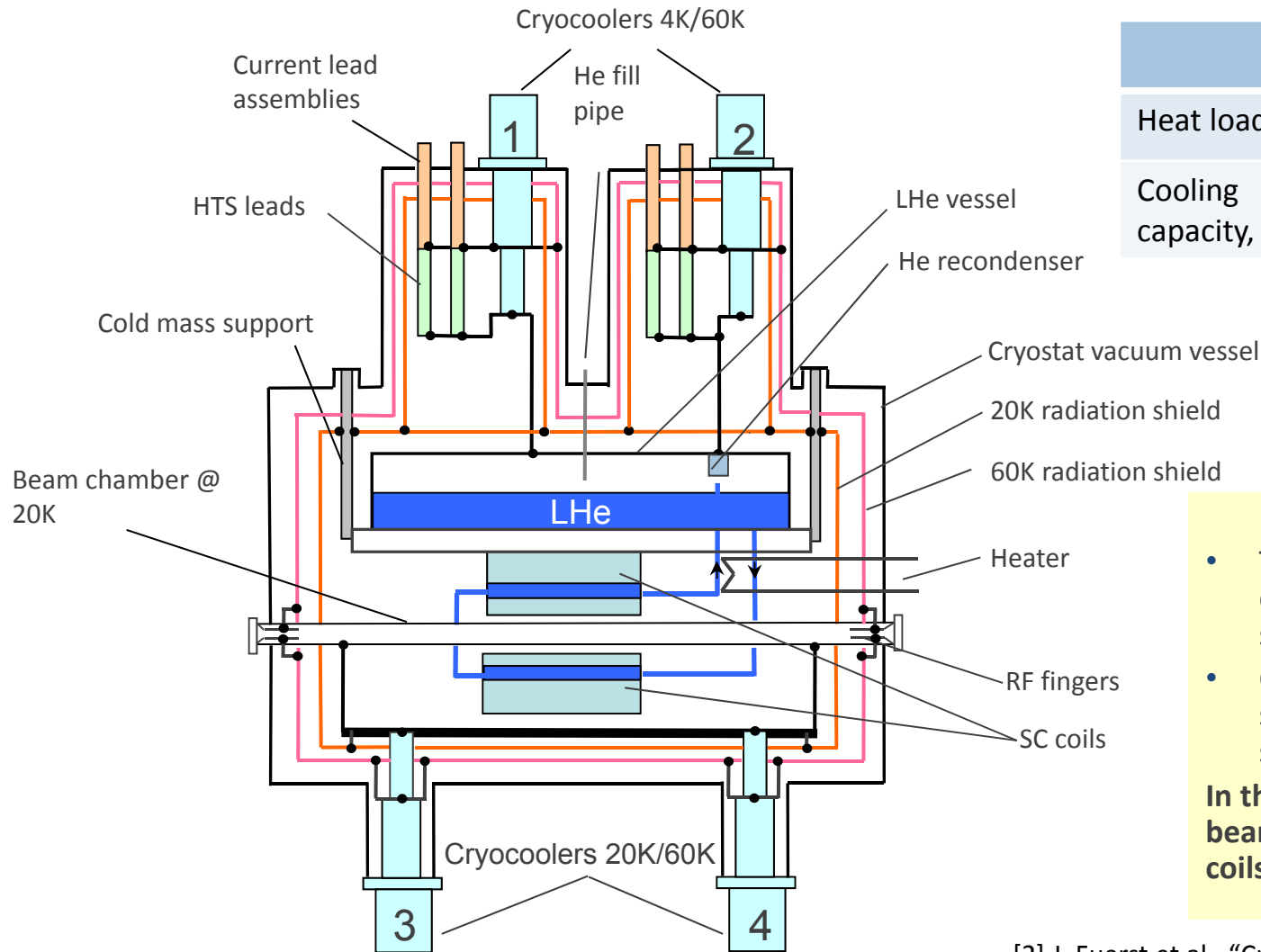
Pros:

- No heating of LHe by beam;
- Cryogen-free system

Cons:

- Temperature difference between the LHe and the coil

SCU0 cooling scheme



	4 K	20 K	60 K
Heat load, W	0.5	10	70
Cooling capacity, W	3	40	224

Conceptual points:

- Thermally insulate beam chamber from the rest of the system.
- Cool the beam chamber separately from the superconducting coils.

In this approach beam heats the beam chamber but not the SC coils!

[2] J. Fuerst et al., "Cryostat design and development for a superconducting undulator for the APS," *Advances in Cryogenic Engineering*, 57A: 901-908, 2012.

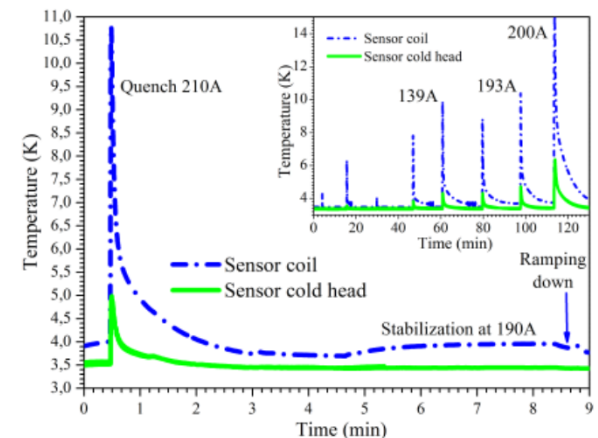
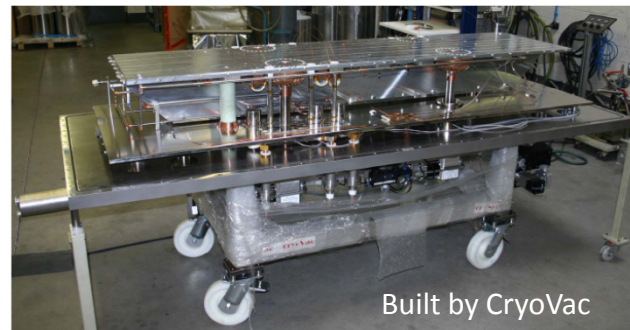
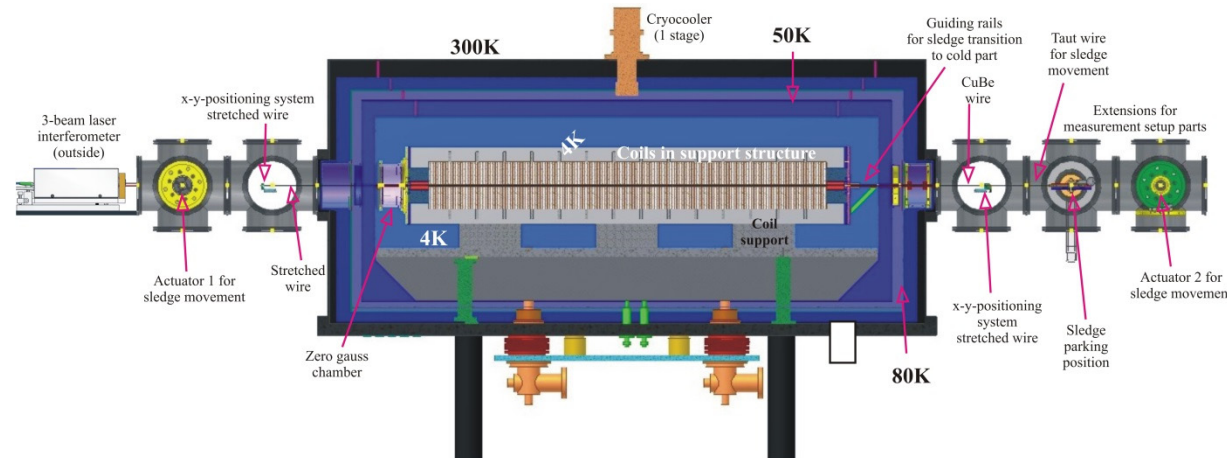
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SCU measurements - Cryogen free test cryostat at ANKA

- CASPER II – ChAracterization Setup for Phase Error Reduction.
- Horizontal cryogen free test of long coils with maximum dimensions 1.5 m in length and 50 cm in diameter.
- Local field measurements with Hall probes. Field integral measurements with stretched wire.

Side view of the cryostat CASPER II



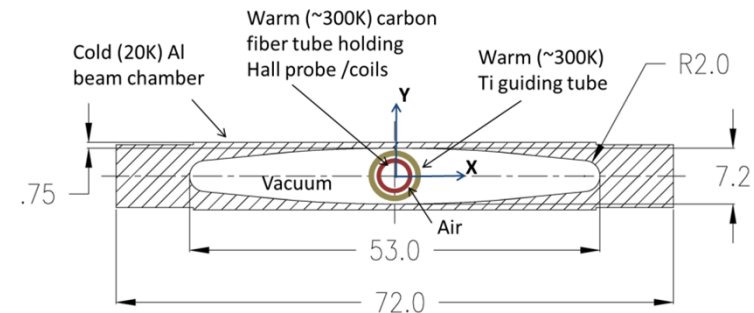
Temperatures over time during quench tests

[1] A. Grau et al., "First Tests with a Local and Integral Magnetic Field Measurement Setup for Conduction Cooled Superconducting Undulator Coils," Proc. Of IPAC2013, p.2138.

Magnetic measurements of SCU0 at APS

- SCU warm-sensor measurement system is based on a concept developed at Budker Institute for characterizing superconducting wigglers.
- Scanning Hall probe:
Specially developed three-sensor Hall probe (attached to carbon fiber tubing and driven by linear stage) to measure B_y and B_x simultaneously and determine the mid-plane field regardless of sensor vertical offset from magnetic mid-plane.
On-the-fly Hall probe measurements (2 cm/s, Δz 0.2 mm, typical z range ± 35 cm) to determine local field errors and phase errors.
- Stretched wire coil
Stretched wire rectangular, delta and 'figure-8' coils to determine static and dynamic 1st and 2nd field integrals.
Rotary stages on upstream end of cryostat as well as on the Z axis linear stage to provide synchronized rotary motion for stretch coils.
Coils can be translated along x axis approximately ± 1 cm to measure integrated multipole components.

SCU warm-sensor concept



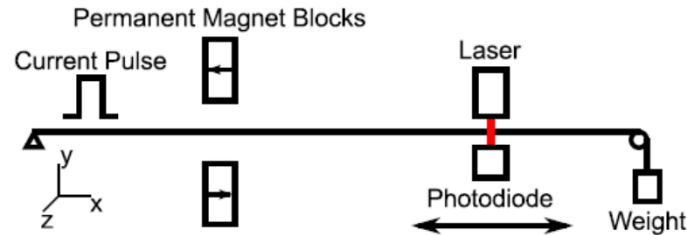
SCU0 horizontal measurement system [1]



[1] C. Doose, M. Kasa, "Magnetic Measurements of the First Superconducting Undulator at the Advanced Photon Source," THPB06, this conference.

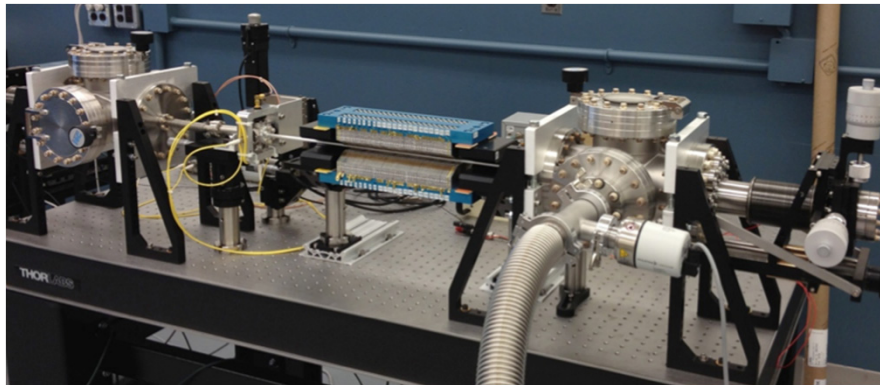
SCU measurements - Pulsed wire at LBNL

Pulsed wire concept



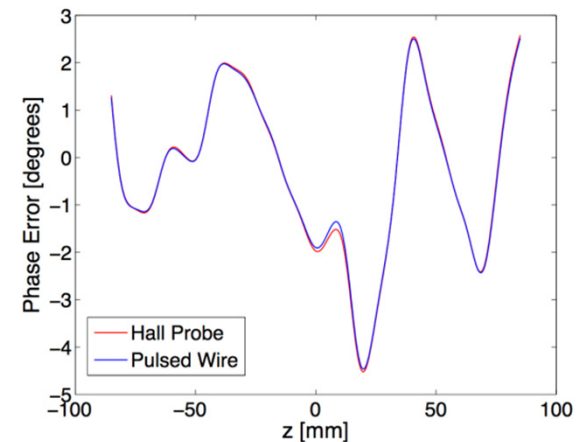
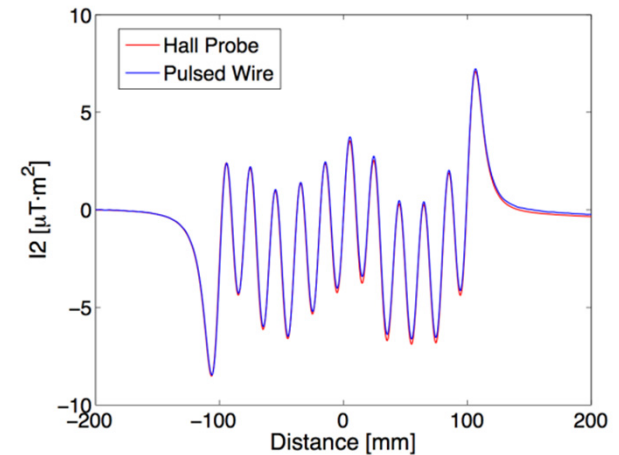
Tensioned wire in the presence of the external magnetic field produced by two permanent magnet blocks. The laser and photodiode pair measure the displacement of the wire as a function of time. In order to change the distance between the sensor and the magnet, the sensor is on a motion stage which is represented by the arrow in the diagram.

Pulsed wire measurement system prototype



D. Arbelaez et al., "A dispersion and pulse width correction algorithm for the pulsed wire method," Nucl. Instrum. Methods. A716 (2013) 62.

Comparison of pulsed wire technique with Hall probe

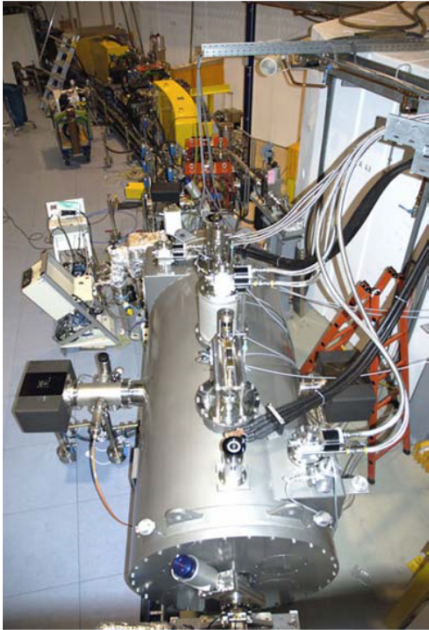


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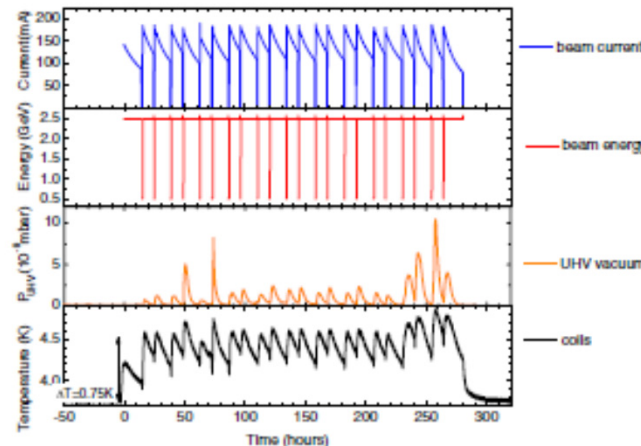
Experience with SCU14 operation at ANKA

Undulator SCU14 in the ANKA storage ring [1]



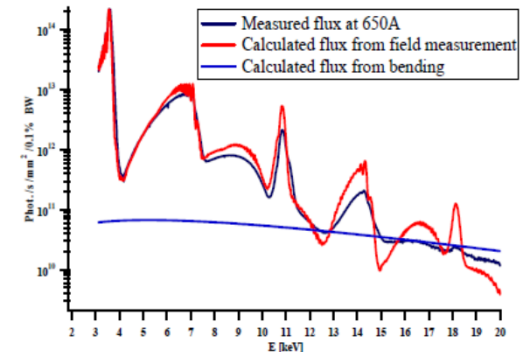
- Built by ACCEL Instruments GmbH
- Installed in 2005
- Intensively used to understand SCU heating mechanism

Typical beam current, beam energy, UHV pressure, SCU coil temperature



- A nonlinear UHV pressure rise with beam current was observed with SCU current off.
- This was attributed to cold chamber electron bombardment effect [3].

SCU14 measured spectrum [2]



Parameter	Value
Period length	14 mm
Number of periods	100
Magnetic gap	8-25 mm
Max. field at 8-mm gap	0.8 T

[1] D. Dolling et al., "Development of Superconducting Undulators at ACCEL," Proc. of SRI06.

[2] R. Rossmanith et al., "A Year's Experience with a Superconducting Undulator in the Storage Ring ANKA," Proc. of EPAC06.

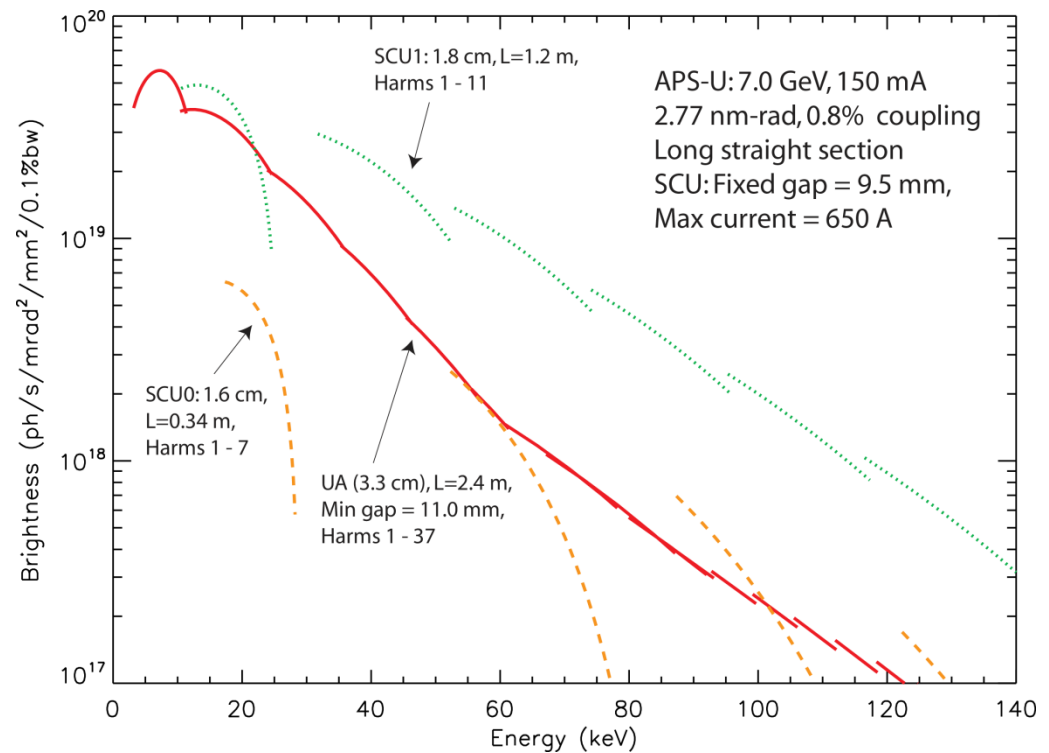
[3] S. Casalbuoni et al., "Beam heat load and pressure rise in a cold vacuum chamber," Phys. Rev. ST Accel. Beams 10 (2007) 093202.

First superconducting undulators at APS

APS superconducting undulator specifications

	Test Undulator SCU0	Prototype Undulator SCU1
Photon energy at 1 st harmonic	20-25 keV	12-25 keV
Undulator period	16 mm	18 mm
Magnetic gap	9.5 mm	9.5 mm
Magnetic length	0.330 m	1.140 m
Cryostat length	2.063 m	2.063 m
Beam stay-clear dimensions	7.0 mm vertical × 36 mm horizontal	7.0 mm vertical × 36 mm horizontal
Superconductor	NbTi	NbTi

SCU0 and SCU1 spectral tuning curves



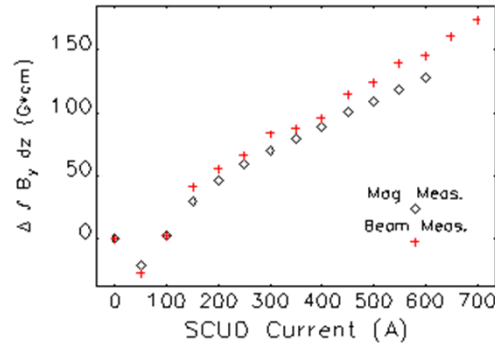
This plot shows the large increases in high-energy flux provided by superconducting devices.

Experience with SCU0 operation at APS

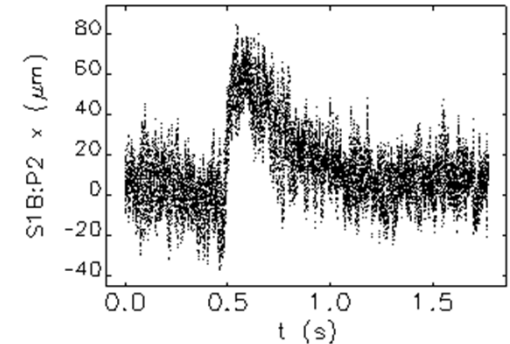
SCU0 in the APS storage ring



Field integral variation
measured with beam



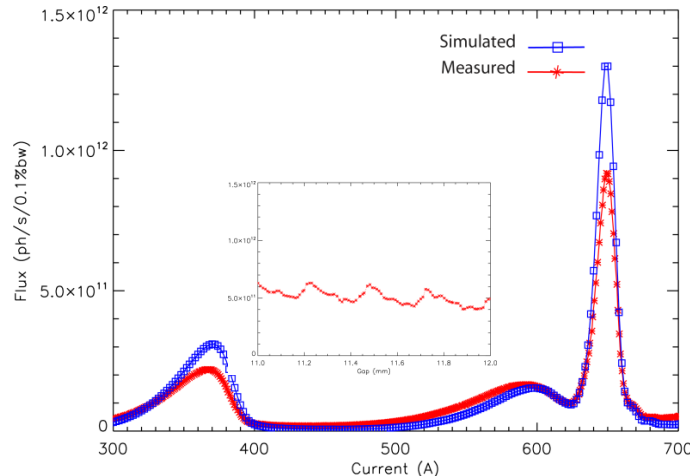
Effect of induced quench
on the beam



SCU0 Performance [1]:

- Designed for operation at 500 A, operates reliably at 650 A
- SCU0 flux at 85 keV is 1.4 times higher than the one of 2.4-m U33 (Undulator A)
- E-beam is not affected by quenches. Didn't quench except of when the e-beam was intentionally dumped
- No loss of He is observed in about 9-month run period

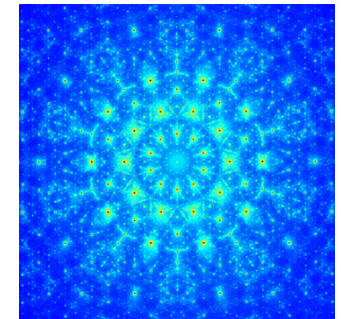
[1] K. Harkay et al., "APS Superconducting Undulator Beam Commissioning Results", WEOAA3, this conference.



Photon flux comparisons at 85 keV.

Main: Simulated and measured SCU0 photon flux.

Inset: Measured photon flux of in-line U33.



First experiments with SCU0. Diffraction pattern from Al-Co-Ni decagonal quasicrystal showing ten-fold symmetry. Data provided by A. Kreyssig and A. Goldman – Iowa State University and Ames Lab.

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Activities in SCUs

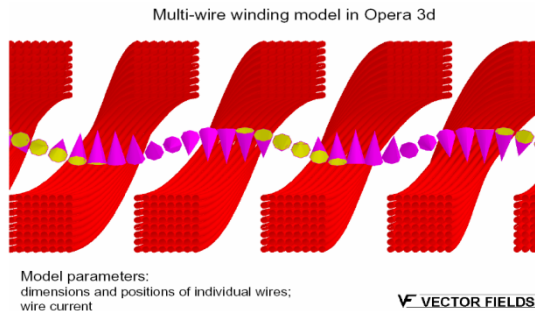
Light Source / Institution	Period (mm)	Pole gap (mm)	K	No. of periods	Super-conductor	Status
NSRRC	15	5.6	1.96	≈132	NbTi	Prototype
SSRF	16	9.5	0.9	10	NbTi	Prototype
ANKA	15	5-16	2.1 max	100.5	NbTi	Being built
DLS	15.5	7.4	1.83	≈129	NbTi	Prototype
APS	16	9.5	1.2	20.5	NbTi	In operation
	18	9.5	1.64	59.5	NbTi	Being built
NGLS/ LBNL	~20	TBD	TBD	~3000	Nb ₃ Sn	Prototype

Scope:

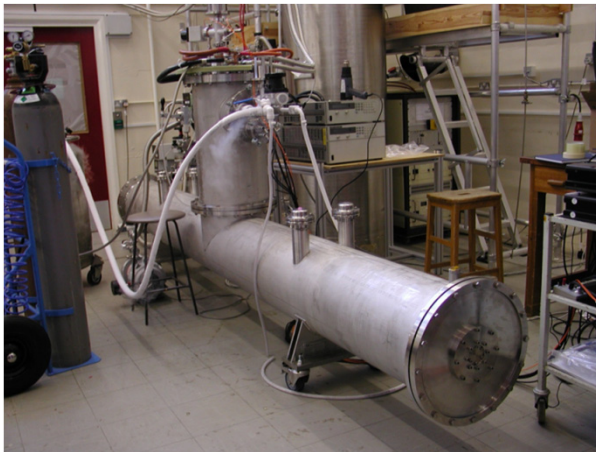
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Helical SCUs

Helical undulator structure



The 4-m long superconducting helical undulator has been built in the UK as a part of the ILC positron source project



D.J. Scott et al., Phys. Rev. Lett. 107, 174803 (2011).

Y. Ivanyushenkov, "Advances in Superconducting Undulators", NA-PAC'13, October 4, 2013

Free electron lasers started in the 1970s with this superconducting undulator:

Superconducting helically wound magnet for the free-electron laser

L. R. Elias and J. M. Madey

High Energy Physics Laboratory, Stanford University, Stanford, California 94305

(Received 12 April 1979; accepted for publication 18 May 1979)

Rev. Sci. Instrum. 50(11), Nov. 1979.

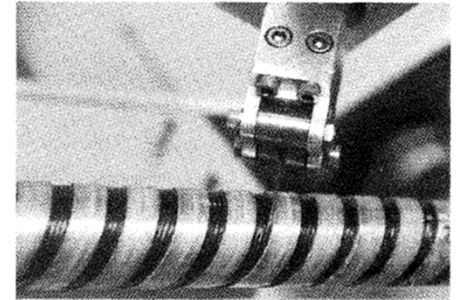


FIG. 5. Wire winding tool and partially completed magnet.

A long line of hybrid undulators in the LCLS Undulator Hall



Picture from SLAC Today, March 30, 2009

<http://today.slac.stanford.edu/feature/2009/lcls-21-undulators.asp>

In principle,
SCUs could
already be
employed in
FELs



Conclusions

- Superconducting undulators produce higher magnetic fields than other undulator technologies.
- Building an SCU is a challenging task. Various solutions have been proposed, tested and implemented.
- Experience of running SCUs at synchrotron light sources confirms that such devices could reliably be operated in storage rings.
- More SCUs are being prototyped, built and proposed.

Acknowledgment and disclaimer

The author thanks S. Casalbuoni, J. Clarke, C.-S. Hwang, S. Prestemon, Z. Zhang, and colleagues from the APS for providing valuable information for this presentation.

It is inevitable with an overview like this that some excellent work that has made valuable contributions to the development of superconducting undulators has been overlooked, but this not intentional.